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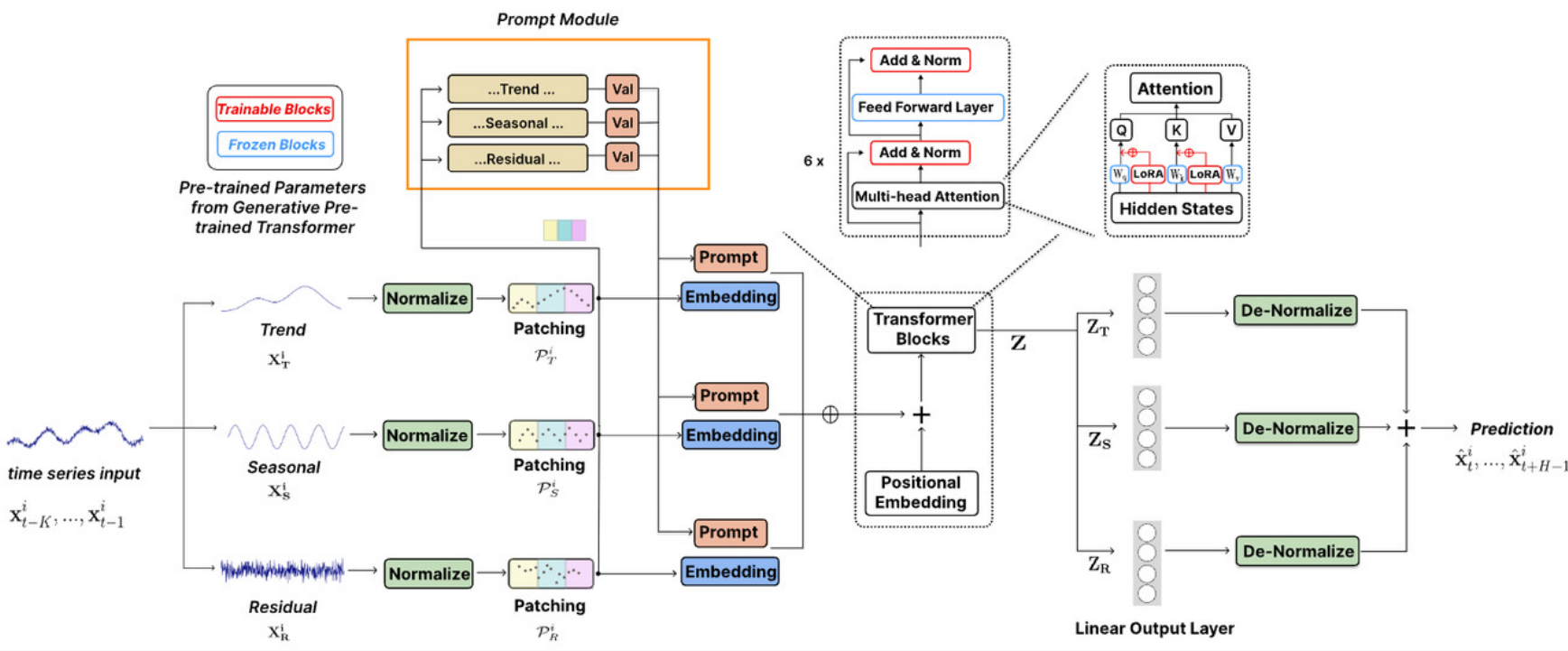
# Time Series Foundation Model - TEMPO: Prompt-based Generative Pre-trained Transformer for Time Series Forecasting

arXiv 2310.04948 | 🤗 Hugging Face Models | License Apache-2.0

The official code for ["TEMPO: Prompt-based Generative Pre-trained Transformer for Time Series Forecasting (ICLR 2024)"].

TEMPO is one of the very first open source **Time Series Foundation Models** for forecasting task v1.0 version.



## ⏳ Upcoming Features

- [✅] Parallel pre-training pipeline
- [] Probabilistic forecasting
- [] Multimodal dataset
- [] Multimodal pre-training script

## 🚀 News

- **Nov 2024**: 🚀 We've published **TimeAGI** on PyPI! Now you can simply `pip install timeagi` to get started and try **TEMPO** by `from tempo.models.TEMPO import TEMPO`. Check out our demo for more details: TimeAGI!
- **Oct 2024**: 🚀 We've streamlined our code structure, enabling users to download the pre-trained model and perform zero-shot inference with a single line of code! Check out our demo for more details. Our model's download count on HuggingFace is now trackable!
- **Jun 2024**: 🚀 We added demos for reproducing zero-shot experiments in Colab. We also added the demo of building the customer dataset and directly do the inference via our pre-trained foundation model: Colab
- **May 2024**: 🚀 TEMPO has launched a GUI-based online demo, allowing users to directly interact with our foundation model!
- **May 2024**: 🚀 TEMPO published the 80M pretrained foundation model in HuggingFace!
- **May 2024**: 🧪 We added the code for pretraining and inference TEMPO models. You can find a pre-training script demo in this folder. We also added a script for the inference demo.
- **Mar 2024**: 📈 Released TETS dataset from S&P 500 used in multimodal experiments in TEMPO.
- **Mar 2024**: 🧪 TEMPO published the project code and the pre-trained checkpoint online!
- **Jan 2024**: 🚀 TEMPO paper get accepted by ICLR!
- **Oct 2023**: 🚀 TEMPO paper released on Arxiv!

## Build the environment

```
conda create -n tempo python=3.8
```

```
conda activate tempo
```

```
pip install timeagi
```

## Script Demo

A streamlining example showing how to perform forecasting using TEMPO:

```
# Third-party library imports
import numpy as np
import torch
from numpy.random import choice
# Local imports
from tempo.models.TEMPO import TEMPO


model = TEMPO.load_pretrained_model(
        device = torch.device('cuda:0' if torch.cuda.is_available() else 'cpu'),
        repo_id = "Melady/TEMPO",
        filename = "TEMPO-80M_v1.pth",
        cache_dir = "./checkpoints/TEMPO_checkpoints"
)

input_data = np.random.rand(336)    # Random input data
with torch.no_grad():
        predicted_values = model.predict(input_data, pred_length=96)
print("Predicted values:")
print(predicted_values)
```

## Demos

### 1. Reproducing zero-shot experiments on ETTh2:

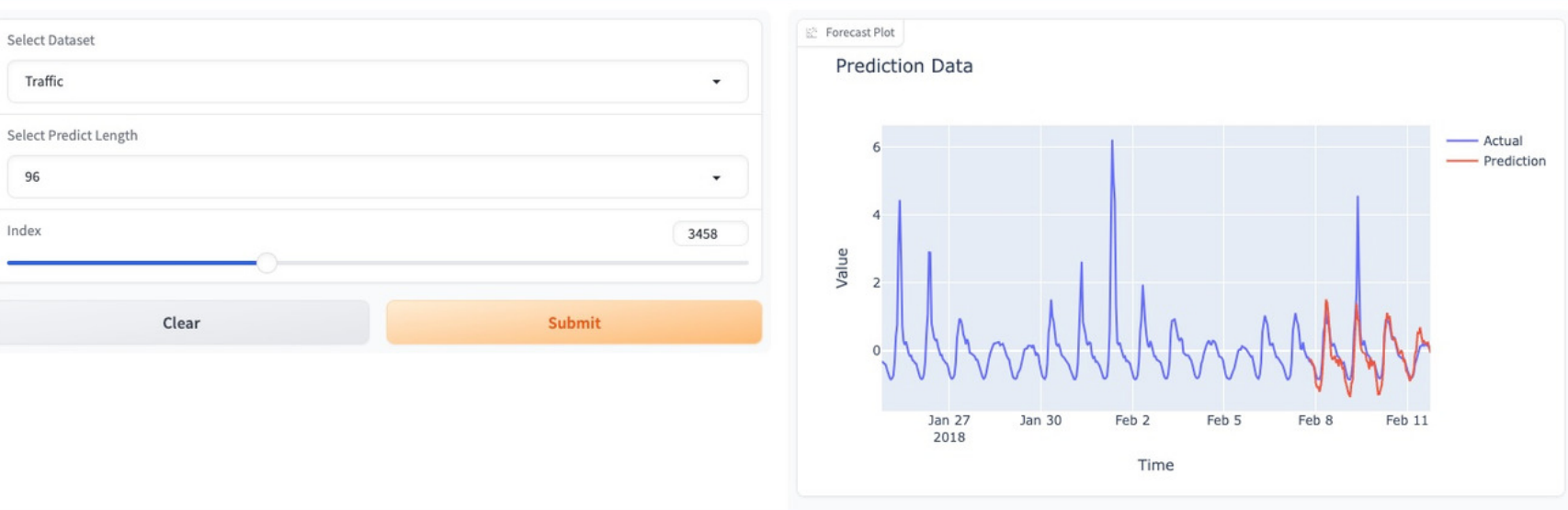
Please try to reproduc the zero-shot experiments on ETTh2 [here on Colab].

### 2. Zero-shot experiments on customer dataset:

We use the following Colab page to show the demo of building the customer dataset and directly do the inference via our pre-trained foundation model: [Colab]

### 3. Online demo:

Please try our foundation model demo [here].


Select Dataset: Traffic
Select Predict Length: 96
Index: 3458

## Practice on your end

We also updated our models on HuggingFace: [Melady/TEMPO].

### Get Data

Download the data from [Google Drive] or [Baidu Drive], and place the downloaded data in the folder `./dataset`. You can also download the STL results from [Google Drive], and place the downloaded data in the folder `./stl`.

### Run TEMPO

### Pre-Training Stage

```
bash [ecl, etth1, etth2, ettm1, ettm2, traffic, weather].sh
```

### Test/ Inference Stage

After training, we can test TEMPO model under the zero-shot setting:

```
bash [ecl, etth1, etth2, ettm1, ettm2, traffic, weather]_test.sh
```

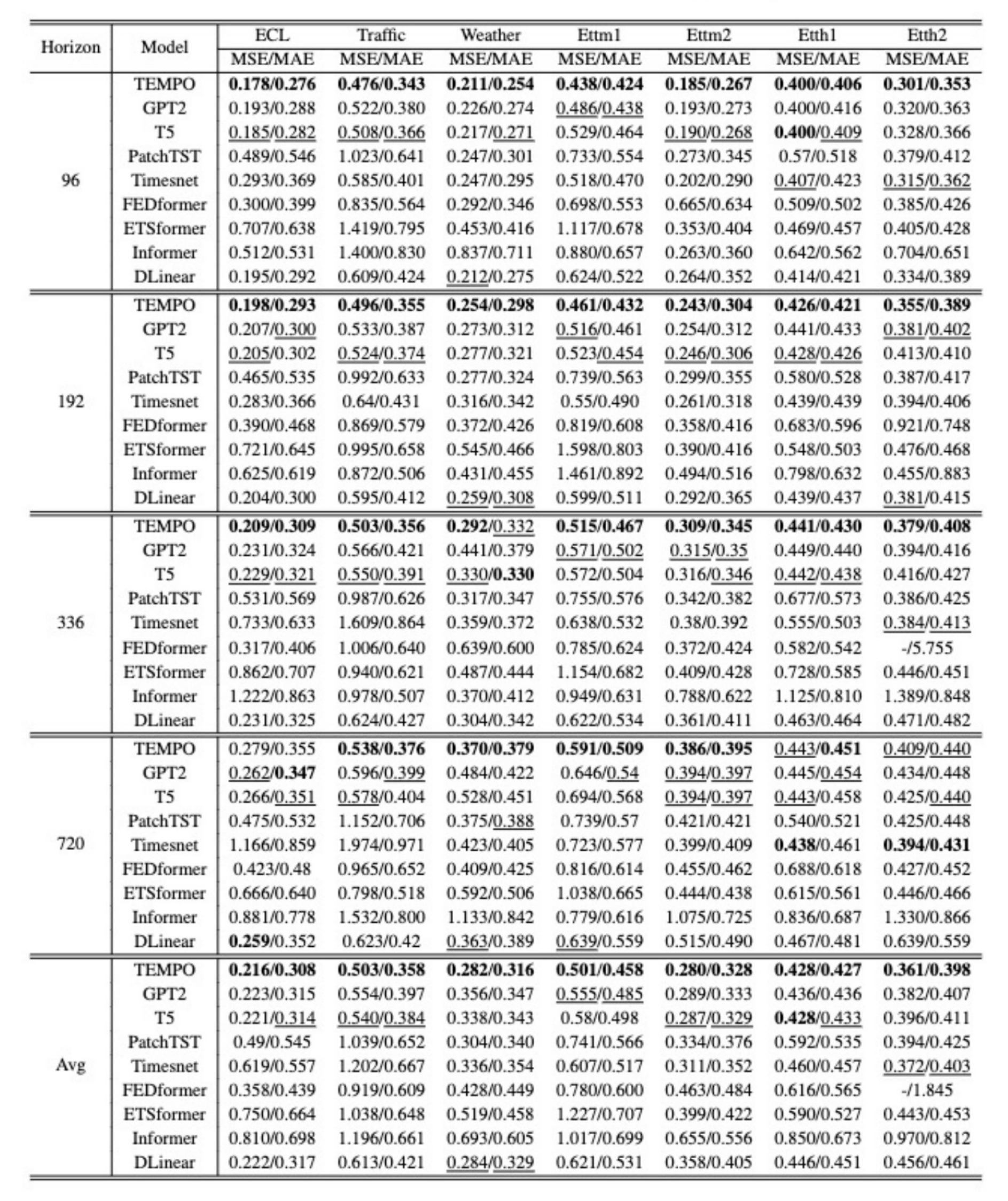


Table 1: Zero-shot of long-term forecasting results on time series benchmark datasets. We use prediction length $O \in \{96, 192, 336, 720\}$. A lower MSE indicates better performance. The best results are marked in bold and the second optimal in underlined, respectively with MSE/MAE.

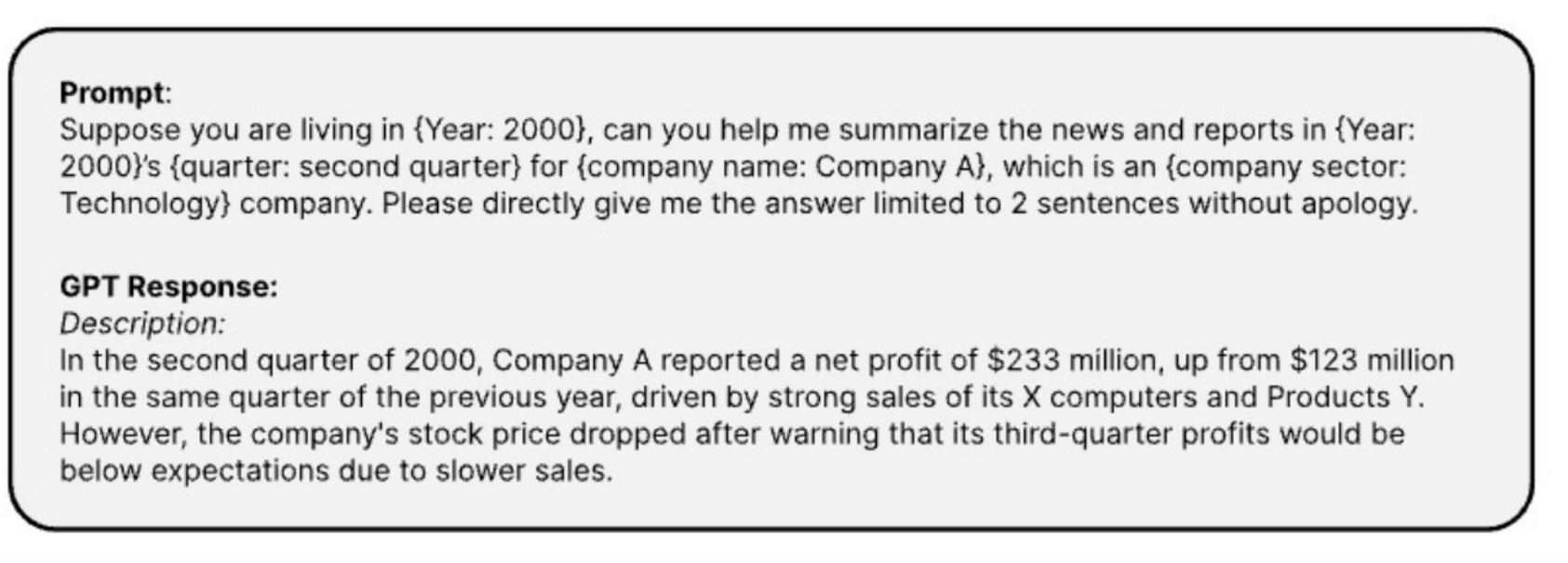
| Horizon | Model | ECL | Traffic | Weather | Ettm1 | Ettm2 | Etth1 | Etth2 |
|---|---|---|---|---|---|---|---|---|
| | | MSE/MAE | MSE/MAE | MSE/MAE | MSE/MAE | MSE/MAE | MSE/MAE | MSE/MAE |
| 96 | TEMPO | **0.178/0.276** | **0.476/0.343** | **0.211/0.254** | **0.438/0.424** | **0.185/0.267** | **0.400/0.406** | **0.301/0.353** |
| | GPT2 | 0.193/0.288 | 0.522/0.380 | 0.226/0.274 | 0.486/0.438 | 0.193/0.273 | 0.400/0.416 | 0.320/0.363 |
| | T5 | 0.185/0.282 | 0.508/0.366 | 0.217/0.271 | 0.529/0.464 | 0.190/0.268 | **0.400**/0.409 | 0.328/0.366 |
| | PatchTST | 0.489/0.546 | 1.023/0.641 | 0.247/0.301 | 0.733/0.554 | 0.273/0.345 | 0.57/0.518 | 0.379/0.412 |
| | Timesnet | 0.293/0.369 | 0.585/0.401 | 0.247/0.295 | 0.518/0.470 | 0.202/0.290 | 0.407/0.423 | 0.315/0.362 |
| | FEDformer | 0.300/0.399 | 0.835/0.564 | 0.292/0.346 | 0.698/0.553 | 0.665/0.634 | 0.509/0.502 | 0.385/0.426 |
| | ETSformer | 0.707/0.638 | 1.419/0.795 | 0.453/0.416 | 1.117/0.678 | 0.353/0.404 | 0.469/0.457 | 0.405/0.428 |
| | Informer | 0.512/0.531 | 1.400/0.830 | 0.837/0.711 | 0.880/0.657 | 0.263/0.360 | 0.642/0.562 | 0.704/0.651 |
| | DLinear | 0.195/0.292 | 0.609/0.424 | 0.212/0.275 | 0.624/0.522 | 0.264/0.352 | 0.414/0.421 | 0.334/0.389 |
| 192 | TEMPO | **0.198/0.293** | **0.496/0.355** | **0.254/0.298** | **0.461/0.432** | **0.243/0.304** | **0.426/0.421** | **0.355/0.389** |
| | GPT2 | 0.207/0.300 | 0.533/0.387 | 0.273/0.312 | 0.516/0.461 | 0.254/0.312 | 0.441/0.433 | 0.381/0.402 |
| | T5 | 0.205/0.302 | 0.524/0.374 | 0.277/0.321 | 0.523/0.454 | 0.246/0.306 | 0.428/0.426 | 0.413/0.410 |
| | PatchTST | 0.465/0.535 | 0.992/0.633 | 0.277/0.324 | 0.739/0.563 | 0.299/0.355 | 0.580/0.528 | 0.387/0.417 |
| | Timesnet | 0.283/0.366 | 0.64/0.431 | 0.316/0.342 | 0.55/0.490 | 0.261/0.318 | 0.439/0.439 | 0.394/0.406 |
| | FEDformer | 0.390/0.468 | 0.869/0.579 | 0.372/0.426 | 0.819/0.608 | 0.358/0.416 | 0.683/0.596 | 0.921/0.748 |
| | ETSformer | 0.721/0.645 | 0.995/0.658 | 0.545/0.466 | 1.598/0.803 | 0.390/0.416 | 0.548/0.503 | 0.476/0.468 |
| | Informer | 0.625/0.619 | 0.872/0.506 | 0.431/0.455 | 1.461/0.892 | 0.494/0.516 | 0.798/0.632 | 0.455/0.883 |
| | DLinear | 0.204/0.300 | 0.595/0.412 | 0.259/0.308 | 0.599/0.511 | 0.292/0.365 | 0.439/0.437 | 0.381/0.415 |
| 336 | TEMPO | **0.209/0.309** | **0.503/0.356** | **0.292**/0.332 | **0.515/0.467** | **0.309/0.345** | **0.441/0.430** | **0.379/0.408** |
| | GPT2 | 0.231/0.324 | 0.566/0.421 | 0.441/0.379 | 0.571/0.502 | 0.315/0.35 | 0.449/0.440 | 0.394/0.416 |
| | T5 | 0.229/0.321 | 0.550/0.391 | 0.330/**0.330** | 0.572/0.504 | 0.316/0.346 | 0.442/0.438 | 0.416/0.427 |
| | PatchTST | 0.531/0.569 | 0.987/0.626 | 0.317/0.347 | 0.755/0.576 | 0.342/0.382 | 0.677/0.573 | 0.386/0.425 |
| | Timesnet | 0.733/0.633 | 1.609/0.864 | 0.359/0.372 | 0.638/0.532 | 0.38/0.392 | 0.555/0.503 | 0.384/0.413 |
| | FEDformer | 0.317/0.406 | 1.006/0.640 | 0.639/0.600 | 0.785/0.624 | 0.372/0.424 | 0.582/0.542 | -/5.755 |
| | ETSformer | 0.862/0.707 | 0.940/0.621 | 0.487/0.444 | 1.154/0.682 | 0.409/0.428 | 0.728/0.585 | 0.446/0.451 |
| | Informer | 1.222/0.863 | 0.978/0.507 | 0.370/0.412 | 0.949/0.631 | 0.788/0.622 | 1.125/0.810 | 1.389/0.848 |
| | DLinear | 0.231/0.325 | 0.624/0.427 | 0.304/0.342 | 0.622/0.534 | 0.361/0.411 | 0.463/0.464 | 0.471/0.482 |
| 720 | TEMPO | 0.279/0.355 | **0.538/0.376** | **0.370/0.379** | **0.591/0.509** | **0.386/0.395** | 0.443/**0.451** | 0.409/0.440 |
| | GPT2 | 0.262/**0.347** | 0.596/0.399 | 0.484/0.422 | 0.646/0.54 | 0.394/0.397 | 0.445/0.454 | 0.434/0.448 |
| | T5 | 0.266/0.351 | 0.578/0.404 | 0.528/0.451 | 0.694/0.568 | 0.394/0.397 | 0.443/0.458 | 0.425/0.440 |
| | PatchTST | 0.475/0.532 | 1.152/0.706 | 0.375/0.388 | 0.739/0.57 | 0.421/0.421 | 0.540/0.521 | 0.425/0.448 |
| | Timesnet | 1.166/0.859 | 1.974/0.971 | 0.423/0.405 | 0.723/0.577 | 0.399/0.409 | **0.438**/0.461 | **0.394/0.431** |
| | FEDformer | 0.423/0.48 | 0.965/0.652 | 0.409/0.425 | 0.816/0.614 | 0.455/0.462 | 0.688/0.618 | 0.427/0.452 |
| | ETSformer | 0.666/0.640 | 0.798/0.518 | 0.592/0.506 | 1.038/0.665 | 0.444/0.438 | 0.615/0.561 | 0.446/0.466 |
| | Informer | 0.881/0.778 | 1.532/0.800 | 1.133/0.842 | 0.779/0.616 | 1.075/0.725 | 0.836/0.687 | 1.330/0.866 |
| | DLinear | **0.259**/0.352 | 0.623/0.42 | 0.363/0.389 | 0.639/0.559 | 0.515/0.490 | 0.467/0.481 | 0.639/0.559 |
| Avg | TEMPO | **0.216/0.308** | **0.503/0.358** | **0.282/0.316** | **0.501/0.458** | **0.280/0.328** | **0.428/0.427** | **0.361/0.398** |
| | GPT2 | 0.223/0.315 | 0.554/0.397 | 0.356/0.347 | 0.555/0.485 | 0.289/0.333 | 0.436/0.436 | 0.382/0.407 |
| | T5 | 0.221/0.314 | 0.540/0.384 | 0.338/0.343 | 0.58/0.498 | 0.287/0.329 | **0.428**/0.433 | 0.396/0.411 |
| | PatchTST | 0.49/0.545 | 1.039/0.652 | 0.304/0.340 | 0.741/0.566 | 0.334/0.376 | 0.592/0.535 | 0.394/0.425 |
| | Timesnet | 0.619/0.557 | 1.202/0.667 | 0.336/0.354 | 0.607/0.517 | 0.311/0.352 | 0.460/0.457 | 0.372/0.403 |
| | FEDformer | 0.358/0.439 | 0.919/0.609 | 0.428/0.449 | 0.780/0.600 | 0.463/0.484 | 0.616/0.565 | -/1.845 |
| | ETSformer | 0.750/0.664 | 1.038/0.648 | 0.519/0.458 | 1.227/0.707 | 0.399/0.422 | 0.590/0.527 | 0.443/0.453 |
| | Informer | 0.810/0.698 | 1.196/0.661 | 0.693/0.605 | 1.017/0.699 | 0.655/0.556 | 0.850/0.673 | 0.970/0.812 |
| | DLinear | 0.222/0.317 | 0.613/0.421 | 0.284/0.329 | 0.621/0.531 | 0.358/0.405 | 0.446/0.451 | 0.456/0.461 |

## Pre-trained Models

You can download the pre-trained model from [Google Drive] and then run the test script for fun.

## TETS dataset

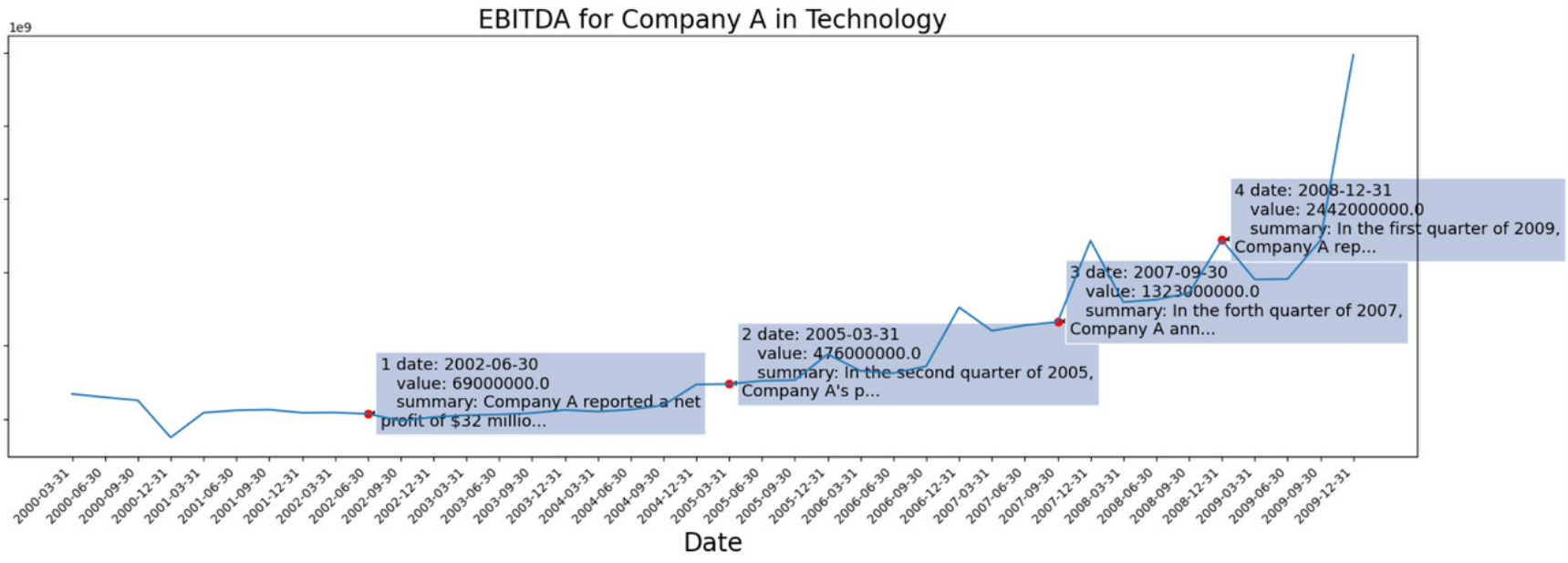
Here is the prompts use to generate the coresponding textual informaton of time series via [OPENAI ChatGPT-3.5 API]



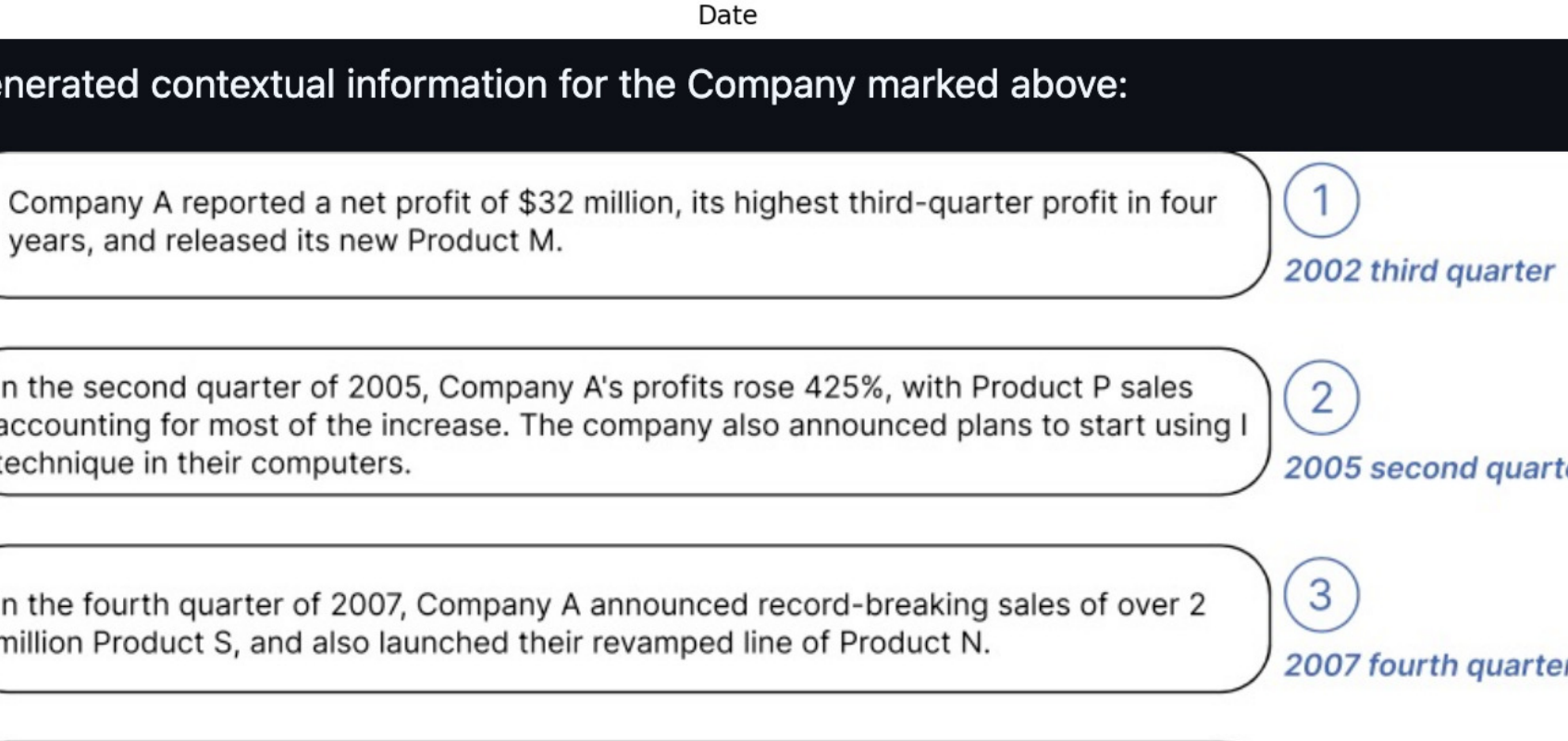
**Prompt**:
Suppose you are living in {Year: 2000}, can you help me summarize the news and reports in {Year: 2000}'s {quarter: second quarter} for {company name: Company A}, which is an {company sector: Technology} company. Please directly give me the answer limited to 2 sentences without apology.

**GPT Response:**
*Description:*
In the second quarter of 2000, Company A reported a net profit of $233 million, up from $123 million in the same quarter of the previous year, driven by strong sales of its X computers and Products Y. However, the company's stock price dropped after warning that its third-quarter profits would be below expectations due to slower sales.

The time series data are come from [S&P 500]. Here is the EBITDA case for one company from the dataset:



Example of generated contextual information for the Company marked above:



Company A reported a net profit of $32 million, its highest third-quarter profit in four years, and released its new Product M. ① *2002 third quarter*

In the second quarter of 2005, Company A's profits rose 425%, with Product P sales accounting for most of the increase. The company also announced plans to start using I technique in their computers. ② *2005 second quarter*

In the fourth quarter of 2007, Company A announced record-breaking sales of over 2 million Product S, and also launched their revamped line of Product N. ③ *2007 fourth quarter*

In the first quarter of 2009, Company A reported a 1% decline in sales and a 17% drop in profits compared to the same period in the previous year, citing the global economic downturn as a contributing factor. The company also announced the release of the U technique software and the new Product IS. ④ *2008 first quarter*

You can download the processed data with text embedding from GPT2 from: [TETS].

## Contact

Feel free to connect DefuCao@USC.EDU / YanLiu.CS@USC.EDU if you're interested in applying TEMPO to your real-world application.

## Cite our work

```
@inproceedings{
cao2024tempo,
title={{TEMPO}: Prompt-based Generative Pre-trained Transformer for Time Series Forecasting},
author={Defu Cao and Furong Jia and Sercan O Arik and Tomas Pfister and Yixiang Zheng and Wen Y
booktitle={The Twelfth International Conference on Learning Representations},
year={2024},
url={https://openreview.net/forum?id=YH5w12OUuU}
}
```

```
@article{
   Jia_Wang_Zheng_Cao_Liu_2024,
   title={GPT4MTS: Prompt-based Large Language Model for Multimodal Time-series Forecasting},
   volume={38},
   url={https://ojs.aaai.org/index.php/AAAI/article/view/30383},
   DOI={10.1609/aaai.v38i21.30383},
   number={21},
   journal={Proceedings of the AAAI Conference on Artificial Intelligence},
   author={Jia, Furong and Wang, Kevin and Zheng, Yixiang and Cao, Defu and Liu, Yan},
   year={2024}, month={Mar.}, pages={23343-23351}
   }
```

### About

The official code for "TEMPO: Prompt-based Generative Pre-trained Transformer for Time Series Forecasting (ICLR 2024)". TEMPO is one of the very first open source Time Series Foundation Models for forecasting task v1.0 version.

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### Releases 2

0.0.1 Latest
2 years ago

+ 1 release

### Deployments 2

pypi last year

### Packages

No packages published

### Contributors 2

**idevede** Defu Cao
**yongchand** Yongchan Hong

### Languages

- Python 95.7%
- Shell 4.3%